

Practical No : 1

Tutorial Title:Write a C++ program to print Pascal Triangle.

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Program:

```
#include <iostream>
using namespace std;
int Fact(int n) {
    int res = 1;
    for (int i = 2; i <= n; i++) {
        res = res * i; }
    return res;
}
int BinoCoef(int a, int b) {
    return Fact(a) / (Fact(b) * Fact(a - b));
}
void printPascal(int l) {
    for (int i = 0; i < l; i++) {
        for (int s = 0; s < l - i - 1; s++) {
            cout << " ";
        }
        for (int j = 0; j <= i; j++) {
            cout << BinoCoef(i, j) << " ";
        }
        cout << endl;
    }
}
int main() {
    int r;
    cout << "Enter number of rows: ";
    cin >> r;
    printPascal(r);
    return 0; }
```

Output:

```
main.cpp Run ▶  
1 #include <iostream>  
2 using namespace std;  
3 int Fact(int n) {  
4     int res = 1;  
5     for (int i = 2; i <= n; i++) {  
6         res = res * i; }  
7     return res;  
8 }  
9 int BinoCoef(int a, int b) {  
10     return Fact(a) / (Fact(b) * Fact(a - b));  
11 }  
12 void printPascal(int l) {  
13     for (int i = 0; i < l; i++) {  
14         for (int s = 0; s < l - i - 1; s++) {  
15             cout << " ";  
16         }  
17         for (int j = 0; j <= i; j++) {  
18             cout << BinoCoef(i, j) << " ";  
19         }  
20         cout << endl;  
21     }  
22 }  
23 int main() {  
24     int r;  
25     cout << "Enter number of rows: ";  
26     cin >> r;  
27     printPascal(r);  
28     return 0;  
29 }
```

Output

Enter number of rows: 5

```
      1  
     1 1  
    1 2 1  
   1 3 3 1  
  1 4 6 4 1
```

=== Code Execution Successful ===